~\OneDrive\Desktop\Shubham2.0\demo.py

```python

import numpy as np

# 1. NumPy Array Creation
arr = np.array([1, 2, 3])
print("1D Array:", arr)

arr = np.array([[1, 2, 3], [4, 5, 6]])
print("2D Array:\n", arr)

arr = np.array((1, 2, 3))
print("Tuple Array:", arr)

print("=" * 40)

# 2. NumPy Slicing and Indexing
arr = np.array([
    [-1, 2, 0, 4],
    [4, -0.5, 0, 0],
    [2.6, 0, 4, 8],
    [3, -7, 4, 2.0]
])

arr2 = arr[:2, ::2]
print("First 2 rows and alternate columns:\n", arr2)

arr3 = arr[[1, 1, 0, 3], [3, 2, 1, 0]]
print("Selected elements:", arr3)

print("=" * 40)

# 3. Arithmetic Operations
a = np.array([[1, 2], [3, 4]])
b = np.array([[4, 3], [2, 1]])

print("Adding 1:\n", a + 1)
print("Subtracting 2:\n", b - 2)
print("Sum of all elements:", a.sum())
print("Array addition:\n", a + b)

print("=" * 40)

# 4. NumPy Data Types
x = np.array([1, 2])
print("Integer dtype:", x.dtype)

x = np.array([1.0, 2.0])
print("Float dtype:", x.dtype)

x = np.array([1, 2], dtype=np.int64)
print("Specified dtype:", x.dtype)
```

```
52
53 print("=" * 40)
54
55 # 5. NumPy Mathematical Functions
56 arr1 = np.array([[4, 7], [2, 6]], dtype=np.float64)
57 arr2 = np.array([[3, 6], [2, 8]], dtype=np.float64)
58
59 print("Addition:\n", np.add(arr1, arr2))
60 print("Sum:", np.sum(arr1))
61 print("Square root:\n", np.sqrt(arr1))
62 print("Transpose:\n", arr1.T)
63
64 print("=" * 40)
65 print("NumPy practice completed successfully!")
66
67
```